A J Patch

FASTED

EDIT/ASM[T.M.] ENHANCEMENTS

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SECTION I - INTRODUCTION

The programs included in this package are a set of modifications to the Radio Shack EDTASM cassette editor. The modifications allow EDTASM to perform its cassette I/O through the JPC Products TC-8 Cassette Interface.

These modifications (or 'patches') allow you to load and store your assembler source code directly through the TC-8. In addition, the object code may be written on tape using a non-standard TC-8 format. The object code can then be loaded into memory using one of the special loaders provided. The final object file is then stored through the TC-8 using the standard 'PUT' command. The object code can be used in the non-standard format but it takes more tape (and time) and loading is not as simple as the TC-8 'GET' command.

Two utility programs are also provided which deal with tape housekeeping functions and give the user the ability to load an object file anywhere in memory.

Throughout this manual the following notation is used: <ENTER> implies pressing the 'ENTER' key on your keyboard and <BREAK> implies pressing the 'BREAK' key on your keyboard. Text enclosed within single quotation marks indicate keyboard entry and information enclosed within double quotation marks must include the double quotes.

There are at least 3 versions of EDTASM produced by Radio Shack to date. In order to proceed with the installation of the FASTED patches you will have to know which version you have purchased. When you first execute Radio Shacks unmodified EDTASM a banner will appear. If the banner is only 1 line and it says VERSION 1.1 then use the name ASM78 in the installation procedure. If the banner is 4 lines long saying VERSION 1.1 then use the name ASM82 in the installation procedure. This version has the origional cassette labeled for Model I/III. Finally, if you have a 1 line banner and it says VERSION 1.2 then install ASM12.

The installation procedure appears in Section IV (page 8). If you are just about to install the FASTED patches you should turn to that section now.

SECTION II - USING FASTED

Insert your new FASTED cassette that was created using the proceedures in SECTION IV. Load your FASTED using the command 'GET "FASTED"'. Type a /<ENTER> after it completes loading. Most of the assembler commands function exactly as described in the EDITOR/ASSEMBLER manual that you received when you purchased it from Radio Shack. None of that information will be described here. The following information describes the changes due to the FASTED patches and the new features.

EDTASM Changes

(1) You MUST specify a <filename> when using the 'W', 'L' and 'A' commands. Absence of a file name will not result in loading the next file on the tape as it would with the unmodified assembler. If you forget the file names you have used, you may go back to the TC-8 operating system and execute the 'GETN' command to list the files on the tape. The prefix character % will appear before all source files and the prefix character & will appear before all object files.

(2) The EDTASM command 'B' (ASM78 & ASM12) or 'Q' (ASM82) command now has four options. They are 'V', 'U', 'T', and <BREAK>.

(A) 'V' - Allows you to verify that the source file you just saved on cassette is good before you leave the assembler. It is similar in purpose to the 'LOAD?' command in the TC-8 operating system. To verify your tape, rewind to a position just before the file and then type

```
'B<ENTER>'
'V'
ENTER FILE NAME (This line is written by FASTED)
```

Type in the name that was used in the 'W' command followed by <ENTER>. The tape will immediately be searched for the filename (assuming the the recorder is in play). There are three possible indications you can get - 'G' indicates that the tape compares exactly with memory; 'B' indicates that the memory has changed since the 'W' command or that the tape is bad; 'C' indicates that there was a checksum error on the tape. In the case of either a 'B' or 'C' you should retry the verify or make a new tape or both. Note - if you do any editing to the file between the 'W' and the 'V' you will probably get a 'B' indication. In order to be sure of a file you should verify IMMEDIATELY after a 'W'.

(B) 'U' causes the TC-8 operating system 'UTIL' to be loaded into memory from your FASTED cassette. The cassette you insert MUST have a copy of UTIL in TC-8 format. Since EDTASM can use all of the available memory (it does not honor MEMSIZE) this command allows you to reload UTIL without having to go back to the TRS-80 tape format (with the attendant connector switching if you have only one drive). This option is not needed if you have the JPC Products EP-38 board with the operating system on EPROM.

EDTASM uses the same memory as the UTIL operating system for its symbol table during the 'A' command. If you have not used the 'A' command and your source does not go that high then this option can be skipped.

(C) The 'T' option is used to terminate FASTED. This will take you out of FASTED and back to MEMSIZE. If you have reloaded UTIL via the 'U' option prior to this then along with UTIL you loaded the special FASTED loader. If you intend to use this loader to load object files created by FASTED then you should specify a MEMSIZE from the 'LOADER' column otherwise use the 'UTIL' column.

	-MEMORY SIZE-		-EXECUTE-	
SYSTEM SIZE	LOADER	UTIL	UTIL	LOADER
16K	30748	31398	31400	30750
32K	47132	47782	47784	47134
48K	63516	64166	64168	63518

(D) The '<BREAK>' option will return you to FASTED if you happened to press 'B' or 'Q' accidentally.

LOADERS

Two programs are provided in order to load your object files into memory. The reason that two are provided is so that if the object code happens to reside in the area of one of the loaders you can use the other one.

One of the programs is loaded along with UTIl from the file created in the installation procedure, the other is in the file 'TAPEUT' which you have also installed on your FASTED tape. TAPEUT occupies the addresses 4350 to 4B00 HEX so the loader in TAPEUT may be used to load anywhere except the addressses 4330 to 4B00. The loader in the file with UTIL occupies the memory addresses x81E to xA70 (x = 7, B or F depending on whether it is a 16K, 32K or 48K system). So, this loader may be used to load in any area except x800 to xA70.

The loaders are fundamentally the same for TC-8 object files. TAPEUT has two additional features which will be described later. To execute TAPEUT just 'GET "TAPEUT"' then type '/<ENTER>'. If TAPEUT is already in memory type 'SYSTEM' followed by '/18176<ENTER>'.

For either loader, select the option - LOAD OBJECT FILE. Place your object tape in the recorder and press the 'PLAY' button on the recorder. Then type in the name of the object file and <ENTER>. The cassette will start, and the asterisks will appear in the upper right hand corner as described in the section "WHAT ARE THE BLINKING ASTERISKS?". After each block is loaded the HEX address following the last BYTE will appear. If you should get a checksum error you will know that the program is sucessfully loaded up to the address which appears. The following block will have also been loaded but some of the program code is in error in that area.

After loading an object file with one of these loaders it is advisable to resave the program using the 'PUT' command through the TC-8. This will eliminate the large amount of tape (and time) used in the inter-record area produced by the assembler format. It will also save memory space since you will not need the loader in memory.

TAPEUT Additional Options

TAPEUT has two additional features. One reads a source file in Radio Shack Format and writes it in TC-8 format. The second feature performs the reverse procedure.

To use the TRS-80 to TC-8 option, connect your recorder to the cassette port in your keyboard. Put your TRS-80 tape in the recorder and press the 'PLAY' button. Respond to the file name request with the name of your TRS-80 source file and press 'ENTER'. The recorder will start and a HEX number counting from 5CF9 (this is where the file is being stored in memory) as the bytes are being read in. At the end it will ask you for the file name you want to use on the TC-8 tape. Connect your recorder to the TC-8 unit #1 output jack, insert a blank tape, place the recorder in 'RECORD/PLAY', type in the new file name and press <ENTER>.

To go from the TC-8 to the TRS-80 just reverse the procedure.

SECTION III - MISCELLANEOUS NOTES

TAPE FORMAT

The TC-8 format for tape consists of 1 or more files, with each file composed of 2 or more records. A file begins with a leader (series of binary zeros) and the end of a file contains a 78H for an object file or two FFH's for a source file.

Blocks begin with a HEX byte 'B3'. The first block in the file is called the file control block. It contains, among other things, the file name. If it is an object file, the file name is preceeded by a '&' and if it is a source file the name is preceeded by a '%'. The next 6 characters are the name specified during the 'W' or 'A' commands of FASTED. Since EDTASM restricts file names to 6 characters, the next two characters are filled with blanks (so that UTIL can read these file control blocks). The next byte is indeterminate, then comes the block check sum byte.

For source files, the entire source text is contained in one block. The block begins with a 'B3' HEX code. The next two bytes contain the start address of the text buffer followed by two bytes containing the final address. Following the addresses is a copy of all the bytes in the source text. The text is terminated by 2 bytes of HEX 'FF'. Then the block checksum appears.

For object files, the blocks are at most 80 HEX bytes long because of the assembler buffer size. Whenever an ORG statement appears, the current block is terminated, regardless of size. The last block of object code will usually be a short record. An object block begins with a HEX 'D3' immediately following the 'B3' beginning of block byte. Next is a byte containing the number of bytes in the block. Following this are 2 bytes specifying the address where the object code is to be loaded. The object code follows terminated by the block checksum. A gap is then placed on the tape in preparation for the next block of object code. The last block of the object file has a HEX '78' code following the 'B3' start of block identifier. Following the '78' code is the transfer address specified on the 'END' statement of your source program.

WHAT ARE THE BLINKING ASTERISKS?

The conventions for the asterisks is different from either the TRS-80 convention or the TC-8 convention. When a read of a file is specified, one asterisk will appear whenever a 'B3' beginning of block is encountered. Thus, during a search on a tape, you can tell if anything is recorded which is detectable by the reading routine. After the beginning of a block is found, the block is searched for the file name. If the proper name is found, a second asterisk is diaplayed. This indicates the proper file has been found. the '%' or '&' are automatically made part of the file name so that you may use the same name for both the object and source files without fear of confusing the program. If the requested file name is not found, the first asterisk will be turned off so that you will know that you have not located the proper file yet.

WHAT ARE THE NUMBERS APPEARING DURING A LOAD?

After a Load, Write or Verify of a source file, a HEX address will appear below the asterisks. This tells you the location of the highest address used to store your source. Don't be alarmed by the fact that the address is slightly different on a Verify and a Write. It differs by exactly 1 byte. This value is presented so that you can determine how much memory you have left for additional source. As you may have discovered, the assembler is completely unpredictable if you have received the 'SYMBOL TABLE OVERFLOW' message. This is because the symbols smash into the end of source text flags. If, during an editing session you wish to determine how much memory you have left, you may do a 'W' command with the recorder turned off. There will be a delay equivalent to the time it takes to write the file, but the highest address used will appear. If you are near the limit, do a 'W' command and save your text before you do the 'A' command because no commands to the assembler can be depended upon after a symbol table overflow. This is an EDTASM bug which carries through into the TC-8 version.

MERGING PROGRAMS

Just as with your unmodified assembler, you may chain programs by doing successive loads. This allows you to have a file of commonly used subroutines or a set of EQUs or whatever, and merge them into your program. Also, several programs can be easily combined into one. With judicous use of loading and deleting statements you may also rearrange blocks of code in your program. One caution to be observed when doing this is that after each chaining operation you should immediately renumber your program using the 'N' command. Also, any intermediate 'END' statements must be deleted before assembling your program becayse the assembly process stops when the first 'END' is encountered.

ACKNOWLEDGEMENTS

The patches to EDTASM were created through a great deal of perseverance and long hours by:

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These patches are Copywrited by JPC Products Company, Inc. Any unauthorized distribution of this software violates not only the law but RIPS OFF Dick Olson. Patches such as FASTED and FASTEX (SCRIPSIT) provide the TC-8 owner with substantial speed and reliability increases over the standard TRS-80 user. They can only be produced if there is adequate compensation for software authors.

SECTION IV - INSTALLATION PROCEDURE

MATERIALS REQUIRED

(1) A copy of your EDTASM program and manual.
(2) A fresh tape.
(3) The installation tape you received with FASTED.

STEPS

(1) Load UTIL as you always have done using the appropriate memory size:

For 16K system - 30748
For 32K system - 47132
For 48K system - 63516

(2) Load EDTASM following the procedures in the EDTASM Manual. DO NOT EXECUTE IT by typing '/<ENTER>' as you normally would. Instead, press the <BREAK> key. If at any time before you execute Step (4) in this procedure you get a 'MEMORY SIZE' message you MUST return to this step. MEMORY SIZE processing changes some of the memory used by EDTASM.

(3) Connect your recorder to Unit #1 on the TC-8. Load the proper loader from the FASTED tape.

For 16K system - GET "LOADER16"<ENTER>
For 32K system - GET "LOADER32"<ENTER>
For 48K system - GET "LOADER48"<ENTER>

After the program has loaded type '/<ENTER>'. The loader will display 3 options. Select option 2 by typing '2'. The loader will respond -

ENTER TC-8 FILE NAME

Remembering which version of EDTASM you have (from the introduction section of this manual) type in the following -

For origional 1.1 - ASM78<ENTER>
For new 1.1 - ASM82<ENTER>
For version 1.2 - ASM12<ENTER>

These overlays each use up about 30 positions on the tape counter so depending on which version you are loading, the load process may take several minutes. After the load is completed, the menu will be displayed again. DO NOT SELECT OPTION 3 AT THIS TIME! This will cause a MEMORY SIZE and you would have to start over. Press the RESET BUTTON near your expansion port on the keyboard to return to command mode.

(4) Remove the installation tape from the recorder. Put your fresh tape in the recorder, advance it beyond the leader, and press the 'PLAY/RECORD' buttons.

(5) Save your new FASTED program accordingly -

```
For ASM78 - PUT "FASTED"4300,5CF9,468A
For ASM82 - PUT "FASTED"4646,6310,4BEA
For ASM12 - PUT "FASTED"4300,5CF9,468A
```

You may verify it by repositioning the tape, press the 'PLAY' button and typing -

```
GET?"FASTED"
```

(6) Press the 'PLAY/RECORD' buttons and save a high speed copy of UTIL and your loader accordingly -

```
For 16K system - PUT "UTIL"781E,7FFF,7AA8
For 32K system - PUT "UTIL"B81E,BFFF,BAA8
For 48K system - PUT "UTIL"F81E,FFFF,FAA8
```

The file in this step MUST be named 'UTIL' in order to reload the operating system when you are done using the assembler. The other file names can be changed if it suits your purposes better.

(7) Remove the tape WITHOUT rewinding it at this time. We will be writing another file on it in step 8. Insert your installation tape in the recorder, press the 'PLAY' button and type -

```
GET "TAPEUT"
```

This will now read in the lower memory loader. Because of the overlays which must be passed over, the tape may advance for some time before 'TAPEUT' is found. Once 'TAPEUT' has been loaded, rewind your installation tape and place it in a safe storage area.

(8) Reinsert the tape you removed in step 7. Press the 'PLAY/RECORD' buttons and type -

```
PUT "TAPEUT"4350,4B00,4700
```

(9) Your tape should now contain three files; FASTED, UTIL, and TAPEUT. These files may be rearranged or backed up using the 'GET' and 'PUT' commands of the TC-8.

(10) Your new high speed Editor/Assembler is now ready for use. ENJOY!